MRC Lecture 4 — Derivation Guide 3

Competitive Binding = Softmax: The Identity

Sai T. Reddy — ETH Zurich / BIIE

1. Motivation

This is the central derivation of Lecture 4 and the **identity moment** of the course. We started Lecture 1 by asking: given a vector of scores, how do we convert them to a probability distribution? The answer was softmax. We now arrive at the same question from biophysics: given a pool of candidate targets with different binding energies, what is the probability that an agent molecule selects each one? The answer is the Boltzmann distribution — which IS softmax.

This guide derives competitive binding from the Boltzmann distribution, writes the result next to the transformer attention equation, and demonstrates with a numerical example that they are the same mathematical object. The identity is visible here but not yet proven unique — that proof comes in Lecture 6.

2. The Competitive Selection Scenario

Physical setup. An agent molecule A (antibody, enzyme, transcription factor, CRISPR guide) encounters a pool of M candidate targets $\{1, 2, \ldots, M\}$ (antigens, substrates, DNA sites). Each pair has a binding free energy:

$$\Delta G_1, \Delta G_2, \ldots, \Delta G_M$$

More negative ΔG_j = stronger binding of agent A to target j.

Question. At thermal equilibrium with all candidates present, what is the probability that agent A binds target j?

Assumption. The agent can bind only one target at a time (mutual exclusion), and binding events occur at thermal equilibrium. These assumptions are standard for molecular recognition at the concentration regime where agents are scarce relative to targets — the biologically relevant regime for most recognition systems.

3. Applying the Boltzmann Distribution

The candidate states are “A bound to target 1”, “A bound to target 2”, …, “A bound to target M”. Each state has energy ΔG_j. From the Boltzmann distribution (Derivation Guide 1):

$$P(A \text{ binds } j) = \frac{\exp(-\Delta G_j / RT)}{\sum_{k=1}^{M} \exp(-\Delta G_k / RT)}$$

Interpretation.

- **Numerator:** the Boltzmann factor for binding to target j. Higher affinity (more negative ΔG_j) → larger numerator.
- **Denominator:** the partition function Z over all candidate targets. This sums over the **entire competitor pool**.
- **Ratio:** the probability that the agent selects target j rather than any of the alternatives.

Why the denominator matters. The probability of binding target j depends not only on ΔG_j but also on the full set of ΔG_k values. A high-affinity target becomes less competitive when stronger competitors are present. This is the **competitive** nature of molecular recognition — and it is captured precisely by the partition function in the denominator.

4. Change of Variables: From Energy to Score

Define the **dimensionless compatibility score**:

$$s_j = -\frac{\Delta G_j}{RT}$$

This is the negated binding energy in units of the thermal energy RT. Interpretation:

- $s_j > 0$ when $\Delta G_j < 0$ (favorable binding)
- $s_j < 0$ when $\Delta G_j > 0$ (unfavorable binding)
- Higher s_j = better compatibility between agent and target j

Substituting into the Boltzmann distribution:

$$P(A \text{ binds } j) = \frac{\exp(s_j)}{\sum_{k=1}^{M} \exp(s_k)}$$

This IS softmax:

$$P(A \text{ binds } j) = \text{softmax}(s)_j$$

where $s = (s_1, s_2, \ldots, s_M)$ is the vector of dimensionless compatibility scores.

5. The Identity with Transformer Attention

The transformer attention equation (Lecture 2, Derivation Guide 2) for the attention weight from query q to key k_j:

$$A_j = \frac{\exp(q \cdot k_j / \tau)}{\sum_{k=1}^{M} \exp(q \cdot k_k / \tau)}$$

where $\tau = \sqrt{d_k}$ in standard transformers (Vaswani et al., 2017).

Writing both equations side by side:

Molecular binding (Boltzmann, 1877)	Transformer attention (Vaswani, 2017)
$P(A \text{ binds } j) = \frac{\exp(-\Delta G_j / RT)}{\sum_k \exp(-\Delta G_k / RT)}$	$A_j = \frac{\exp(q \cdot k_j / \tau)}{\sum_k \exp(q \cdot k_k / \tau)}$

Under the identification:

$$s_j = -\frac{\Delta G_j}{RT} = \frac{q \cdot k_j}{\tau}$$

These are the same equation.

- The dot product $q \cdot k_j$ maps to the negated binding energy $-\Delta G_j$.
- The attention temperature τ maps to the thermal energy RT.
- The attention weight A_j maps to the Boltzmann selection probability.
- The softmax denominator maps to the partition function.

What this means. What molecular biology calls a *binding probability* and what machine learning calls an *attention weight* are the same mathematical object. The transformer attention mechanism — the core operation of modern AI — is the Boltzmann distribution applied to molecular compatibility scores.

Lecture 6 will prove this is the **unique** function satisfying five axioms of competitive selection: no other function can sit in this equation. For now, we observe the identity and demonstrate it numerically.

6. Worked Example: Three-Antigen Selection

Problem. An antibody A encounters three candidate antigens with binding free energies (at $T=310$ K, where $RT=0.616$ kcal/mol):

Antigen	ΔG (kcal/mol)
1	-10.0 (tight binder)
2	-8.0 (moderate)
3	-6.0 (weak)

Compute the probability that A selects each antigen.

Step 1: Compute dimensionless scores.

$$s_1=-\Delta G_1/RT=-(-10.0)/0.616=16.23$$
$$s_2=-(-8.0)/0.616=12.99$$
$$s_3=-(-6.0)/0.616=9.74$$

Step 2: Exponentiate.

$$\exp(s_1)=\exp(16.23)=1.11\times10^7$$
$$\exp(s_2)=\exp(12.99)=4.38\times10^5$$
$$\exp(s_3)=\exp(9.74)=1.70\times10^4$$

Step 3: Partition function.

$$Z=1.11\times10^7+4.38\times10^5+1.70\times10^4=1.155\times10^7$$

Step 4: Probabilities.

$$P(\text{bind }1)=\frac{1.11\times10^7}{1.155\times10^7}=0.961$$
$$P(\text{bind }2)=\frac{4.38\times10^5}{1.155\times10^7}=0.038$$
$$P(\text{bind }3)=\frac{1.70\times10^4}{1.155\times10^7}=0.0015$$

Verify: $0.961+0.038+0.0015=1.0005\approx1.000$ ✓ (small rounding)

Interpretation.

Antigen 1 ($\Delta G=-10$ kcal/mol) captures 96.1% of the binding probability. Antigen 2 gets 3.8%, and antigen 3 barely registers at 0.15%. A 2 kcal/mol difference in binding energy (between antigens 1 and 2, about 3.2 RT) translates to a 25-fold preference. A 4 kcal/mol difference (between antigens 1 and 3, about 6.5 RT) translates to a ~640-fold preference.

The scores vector $(16.23, 12.99, 9.74)$ fed into softmax gives the probability vector $(0.961, 0.038, 0.0015)$. This is exactly the same computation a transformer performs when computing attention weights — just with dot-product scores instead of negated binding energies.

---

7. Effect of Temperature (Pool of Same Three Antigens)

The same three antigens at three different temperatures. This demonstrates the role of the temperature parameter — which is exactly the role $\tau=\sqrt{d_k}$ plays in transformer attention.

At T = 310 K (physiological, RT = 0.616 kcal/mol):

From the computation above: $(P_1, P_2, P_3) = (0.961, 0.038, 0.0015)$. Sharp selection.

At T = 620 K (hypothetical high temperature, RT = 1.232 kcal/mol):

$$s_1 = 10.0/1.232 = 8.12, s_2 = 8.0/1.232 = 6.49, s_3 = 6.0/1.232 = 4.87$$
$$\exp(s_1) = 3368, \exp(s_2) = 659.9, \exp(s_3) = 130.3$$
$$Z = 4158$$
$$P_1 = 0.810, P_2 = 0.159, P_3 = 0.031$$

At higher temperature: the distribution flattens. Antigen 1 still dominates, but antigens 2 and 3 become meaningfully populated.

At T = 155 K (hypothetical low temperature, RT = 0.308 kcal/mol):

$$s_1 = 10.0/0.308 = 32.47, s_2 = 25.97, s_3 = 19.48$$
$$\exp(s_1) = 1.28 \times 10^{14}, \exp(s_2) = 1.92 \times 10^{11}, \exp(s_3) = 2.89 \times 10^{8}$$
$$P_1 = 0.99985, P_2 = 0.00015, P_3 \approx 10^{-6}$$

At low temperature: the distribution becomes nearly deterministic — antigen 1 captures essentially all the probability.

Summary of temperature effects:

Temperature	RT (kcal/mol)	P(1)	P(2)	P(3)	Character
155 K	0.308	0.9999	0.0001	~0	Nearly deterministic
310 K	0.616	0.961	0.038	0.002	Sharp but probabilistic
620 K	1.232	0.810	0.159	0.031	More diffuse

This is the same behavior as the temperature regimes of softmax from Lecture 1. Low temperature → sharp selection; high temperature → diffuse selection. The physical temperature RT and the machine learning temperature τ play identical mathematical roles.

8. What the Identity Does and Does Not Establish

What is established by this derivation:

1. The Boltzmann distribution is softmax applied to dimensionless compatibility scores.
2. Competitive molecular binding at thermal equilibrium follows the softmax form.
3. The mathematical structure of the Boltzmann selection probability is identical to transformer attention.

What is NOT yet established (but will be in later lectures):

1. **That softmax is the unique function** satisfying physically reasonable axioms of selection. Multiple functions could in principle produce probability distributions; why must it be exponential? The uniqueness proof is Lecture 6 (Luce's theorem).
2. **That the binding energy decomposes as a dot product.** The derivation above treats ΔG_j as a scalar for each target. The claim that $\Delta G_j = -q \cdot k_j$ (bilinear decomposition) is Lecture 5. For nucleic acids (given bilinearity), this is exact. For proteins (emergent bilinearity), it is learned.
3. **That this identity prescribes a complete neural architecture.** Given the Boltzmann-softmax equivalence and the bilinear energy decomposition, the full SFM architecture — dual encoders, InfoNCE, symmetric loss, learned temperature — follows. This is Lecture 7.

The identity derived here is the **entry point** to the MRC framework. Each subsequent lecture fills in one of the remaining pieces.

9. Generalization Across Molecular Recognition Systems

The derivation made no assumption specific to antibodies. It applies to any molecular recognition system at thermal equilibrium:

System	Agent	Target pool	Selection probability
Antibody-antigen	Antibody A	Antigens $\{1,\dots,M\}$	Probability A binds each antigen
TCR-pMHC	T-cell receptor	Peptide-MHC complexes	Probability TCR recognizes each pMHC
TF-DNA	Transcription factor	Genomic binding sites	Probability TF binds each site
Enzyme-substrate	Enzyme	Substrates	Probability enzyme acts on each substrate
miRNA-mRNA	microRNA	Target sites in mRNAs	Probability miRNA targets each site
CRISPR-DNA	Guide RNA	Genomic loci	Probability gRNA directs Cas to each locus
Drug-target	Drug molecule	Protein targets	Probability drug binds each target

In every case: competitive selection from a pool, thermal equilibrium, Boltzmann distribution. **One equation, seven systems.** Lecture 8 will show that this extends to at least ten biological domains, with six already prototyped as SFMs (Reddy, *bioRxiv*, 2026 — the vibe-coding paper).

10. The Historical Convergence

The identity demonstrated in this derivation has been reached independently by six fields over 150 years:

Year	Discipline	Discoverer	Formulation
1877	Statistical mechanics	Boltzmann	$P(E) \propto \exp(-E/k_B T)$ (derived)
1948	Information theory	Shannon	Maximum entropy distribution
1957	Information theory	Jaynes	Uniquely rational inference under incomplete information
1959	Mathematical psychology	Luce	Unique function satisfying five axioms of rational choice
1974	Econometrics	McFadden	Discrete choice theory (Nobel Prize 2000)
2017	Machine learning	Vaswani et al.	Scaled dot-product attention (transformer)
2026	Molecular biology	Reddy	The convergence equation — identity formalized

What each field called it:

- Boltzmann: **partition function**
- Shannon/Jaynes: **maximum entropy distribution**
- Luce: **choice probability**
- Vaswani: **softmax attention**

They are the same mathematical object. Lecture 6 proves this is not coincidence — the five axioms each field independently identified converge on a single functional form.

11. Summary

Concept	Result	Significance
Competitive Boltzmann	$P(j)=\exp(-\Delta G_j/RT)/Z$	Multi-candidate selection at equilibrium
Dimensionless score	$s_j=-\Delta G_j/RT$	Energy in thermal units
Identity	$P(j)=\mathrm{softmax}(s)_j$	Boltzmann IS softmax
Attention correspondence	$-\Delta G_j \leftrightarrow q \cdot k_j$; $RT \leftrightarrow \tau$	One equation, two fields

Concept	Result	Significance
Partition function	$Z = \sum_k \exp(s_k)$	The softmax denominator
Temperature effects	Sharp ↔ diffuse selection	Same physics as ML temperature
Six independent derivations	Boltzmann → Shannon → Jaynes → Luce → McFadden → Vaswani → Reddy	150 years, one function

12. Checkpoint Questions

Q1. In the worked example (§6), the difference in binding energy between antigens 1 and 2 is 2 kcal/mol (about 3.2 RT). The probability ratio is 0.961/0.038 ≈ 25. Verify that this ratio equals $\exp(s_1 - s_2)$ and explain the shift-invariance property.

Answer: $\exp(s_1 - s_2) = \exp(16.23 - 12.99) = \exp(3.24) = 25.5$ ✓ (matches within rounding). The ratio depends only on the score difference, not on the absolute scores. Adding a constant C to all scores ($s_j \to s_j + C$) would multiply every numerator and every denominator term by $\exp(C)$, which cancels — leaving the probabilities unchanged. This shift invariance is the **IIA property** (Independence of Irrelevant Alternatives) — the relative preference between two targets depends only on their scores, not on absolute values or on the presence of other candidates. Lecture 6 proves this property uniquely selects the exponential form.

Q2. A student argues: "At very low temperature, the probability of the best-binding target approaches 1. But the physics of binding happens at 310 K — why do we even need softmax? Why not just pick the argmax?"

Answer: At physiological temperature (310 K, $RT = 0.616$ kcal/mol), the thermal energy is substantial. Small energy differences (1–3 kcal/mol, or 2–5 RT) are typical for molecular discrimination, and these produce graded — not deterministic — selection. In the worked example, antigen 1 was 96% probable, not 100%: 4% of the time, the antibody binds a different (weaker) target due to thermal fluctuations. This is the biological basis of cross-reactivity and off-target effects. Using argmax would eliminate this stochasticity and give a false picture of molecular selectivity. The **soft, probabilistic** selection captured by softmax is the correct physical description. This is why Axiom 1 (positivity — every candidate has non-zero probability) is in Luce's list of five.

Q3. In the transformer, the temperature is $\tau = \sqrt{d_k}$ — determined by the embedding dimension. In physics, the temperature is RT — determined by the physical temperature of the system. In SFMs, the temperature is learned. Why is learned temperature the correct choice, given the identity?

Answer: The identity tells us $s_j = -\Delta G_j / RT = q \cdot k_j / \tau$. If we want the contrastive score $q \cdot k_j$ to recover the physical binding energy, then τ must match RT — up to a unit conversion between embedding space (arbitrary units) and kcal/mol (physical units). The embedding units are determined by encoder architecture, L2 normalization, and training dynamics, so τ cannot be known a priori. A learned temperature allows the model to discover the correct unit conversion. CLIP (Radford et al., 2021) used learned temperature empirically; SFMs adopt it because the physics requires it. Lecture 7 gives the full argument; Lecture 10 (future work) uses learned τ to recover physical temperature from the embedding geometry.

Q4. The derivation computed $P(\text{bind } j)$ for each target in the pool. But the pool consists of specific antigens in a specific experiment. If a new antigen is added to the pool, does the Boltzmann probability for the original antigens change?

Answer: Yes — the partition function Z now includes the new term $\exp(s_{\text{new}})$, so every $P(j)$ is reduced. But the **ratio** between any two original targets is unchanged: $P(i)/P(j) = \exp(s_i - s_j)$, which doesn't depend on the new target's score. This is exactly the IIA property: the relative preference between original targets is independent of the presence or absence of other candidates. However, the absolute probabilities change because probabilities must sum to 1 over the full pool. In SFM training, this is why the in-batch competitor pool matters — the batch defines the "competition" for the

InfoNCE denominator. Different pool compositions give different absolute selection probabilities but preserve relative rankings.

13. Connection to Future Lectures

- **Lecture 5:** The binding energy ΔG_j is shown to decompose as a dot product: $-\Delta G_j = q \cdot k_j$ (given bilinearity for nucleic acids, emergent bilinearity for proteins). Substituting this into the Boltzmann distribution derived here gives the attention equation directly.
- **Lecture 6:** The five axioms that uniquely select the exponential form are stated. The identity demonstrated here is proven to be inevitable.
- **Lecture 7:** The five architecture conditions combine with the convergence equation to prescribe a complete neural architecture — the SFM. Dual encoders, InfoNCE, symmetric loss, learned temperature all follow from the identity derived here.
- **Lecture 8 onward:** The identity is applied across ten molecular recognition domains. Six are already prototyped (CALM, tSFM, crisprSFM, miR-SFM, mhcSFM, eSFM, dtSFM). Students build their own SFM using this identity as the foundation.